

Day 4 Task Solution

October 23, 2025

1 General Process Overview

This workflow automates the processing of invoices received as PDF files. The goal is to extract key information from potentially differently formatted invoices, update a database (Google Sheet), and notify an internal team (e.g., billing). It leverages AI for information extraction because the invoice format isn't consistent, making simple rule-based extraction unreliable.

The overall flow is:

1. **Trigger:** A new invoice (PDF) is added to a specific Google Drive folder.
2. **Download:** The PDF file is downloaded from Google Drive to access its content.
3. **Extract Text:** The text content is extracted from the PDF file.
4. **Extract Information (AI):** An AI model reads the extracted text and pulls out predefined fields (e.g., invoice number, client name, amount, dates).
5. **Update Database:** The extracted information is added as a new row in a Google Sheet acting as an invoice database.
6. **Craft Email (AI):** Another AI model drafts an email notification for the internal billing team, summarizing the key invoice details and linking to the database.
7. **Send Email:** The drafted email is sent to the designated internal billing email address.

2 Critical Node Explanations

Here's a breakdown of the important configurations and functions of each node in the workflow:

2.1 Google Drive Trigger (On changes involving a specific folder)

- **Critical Function:** Monitors a designated folder in Google Drive.
- **Configuration:**
 - **Folder:** Set to the specific folder where new invoices will be placed (e.g., "Projects" folder in the example).
 - **Watch For:** Configured to `File Created` to trigger only when a new file appears.
 - **Credential:** Requires authenticated Google Drive access.
- **Output:** Provides metadata about the new file, crucially including its `File ID`.

2.2 Google Drive Node (Download File)

- **Critical Function:** Retrieves the actual file content based on the ID from the trigger.
- **Configuration:**
 - **File ID:** Set dynamically using an expression (`{{ $json.id }}` or similar) that pulls the File ID from the *previous* Google Drive Trigger node's output. This ensures it always downloads the *new* file.
 - **Credential:** Requires authenticated Google Drive access.
- **Output:** Provides the file content as Binary Data, which is needed for text extraction.

2.3 Extract from File Node (Extract from PDF)

- **Critical Function:** Reads the binary data of the PDF and converts it into plain text.
- **Configuration:**
 - **Input Binary Field:** Set to data (or the field name containing the binary output from the "Download File" node).
- **Output:** Plain text content of the invoice.

2.4 AI Information Extractor Node

- **Critical Function:** Uses an AI model (like Gemini Flash) to parse the unstructured text and extract specific, predefined pieces of information.
- **Configuration:**
 - **Chat Model:** Needs an AI model connected (e.g., Gemini, OpenAI).
 - **Text to Extract From:** Set dynamically using an expression to pull the text output from the "Extract from File" node.
 - **Attributes to Extract:** Define *each* piece of information needed (e.g., invoice number, client name, total amount, due date). For each attribute:
 - ✱ Give it a clear Name.
 - ✱ Provide a concise Description telling the AI what this piece of information is (e.g., "The number of the invoice").
 - ✱ Set the expected Data Type (String, Number, Date). Crucial for dates and amounts if specific formatting is needed later.
 - ✱ Mark as Required if the information must be present.
- **Output:** A structured JSON object containing the extracted fields as key-value pairs.

2.5 Google Sheets Node (Append Row) - Update Database

- **Critical Function:** Adds the extracted invoice data as a new row in the designated spreadsheet.
- **Configuration:**
 - **Document:** Select the correct Google Sheet file (e.g., "Invoice DB").
 - **Sheet:** Select the specific sheet within the file (e.g., "Sheet1").

- **Columns:** Map *each* column in your Google Sheet to the corresponding output field from the "AI Information Extractor" node using dynamic expressions (e.g., map the invoice number sheet column to `{{ $json.output["invoice number"] }}`). Ensure the mapping is correct.
- **Credential:** Requires authenticated Google Sheets access.
- **Output:** Confirmation that the row was added.

2.6 AI Model Node (OpenAI Message / Similar) - Create Email

- **Critical Function:** Drafts the notification email content using an AI model.
- **Configuration:**
 - **Chat Model:** Connect an AI model (e.g., GPT-4o Mini).
 - **System Prompt:** Instructs the AI on its role (e.g., "You are an email expert for Green Grass Corp named Greeny...") and the desired email structure/content (e.g., "Notify the internal billing team...", "Include a link to the database..."). Can include static elements like the database link.
 - **User Prompt:** Provides the *dynamic* data for *this specific* invoice, pulled from the "AI Information Extractor" output using expressions (e.g., Invoice Number: `{{...}}`, Client Name: `{{...}}`, Due Date: `{{...}}`). Only include the fields relevant for the email.
 - **Output Content as JSON: **Crucially important.**** Enable this option and define the JSON structure (e.g., `{"subject": "...", "body": "..."}`) so the output separates the subject and body for the next node. Use Chat GPT to help generate this JSON structure if needed.
- **Output:** A JSON object containing the `subject` and `body` of the email.

2.7 Gmail Node (Send Email)

- **Critical Function:** Sends the email drafted by the AI node.
- **Configuration:**
 - **To:** Set to the static internal email address (e.g., `billing@example.com`).
 - **Subject:** Set dynamically using an expression to pull the `subject` field from the *previous* "Create Email" node's JSON output (e.g., `{{ $json.output.subject }}`).
 - **Message:** Set dynamically using an expression to pull the `body` field from the *previous* "Create Email" node's JSON output (e.g., `{{ $json.output.body }}`).
 - **Email Type:** Typically Text.
 - **Append n8n Attribution:** Usually turned off for professional emails.
 - **Credential:** Requires authenticated Gmail access.
- **Output:** Confirmation that the email was sent.

This workflow demonstrates how to handle variable inputs (differently formatted PDFs) using AI and integrate multiple services (Google Drive, AI Models, Google Sheets, Gmail) for a complete automated process.